

Modeling the barriers toward the growth of higher education institutions

A total interpretive structural modeling approach

Faisal Talib

Department of Mechanical Engineering, Zakir Husain College of Engineering and Technology, Aligarh Muslim University, Aligarh, India, and

Zillur Rahman

Department of Management Studies, Indian Institute of Technology Roorkee, Roorkee, India

Barriers
affecting HEIs
in India

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Abstract

Purpose – In recent years, sustainable issues have gained increasing importance in the higher education sector across the country. To remain competitive, higher education institutions (HEIs) have to engage themselves in providing world-class opportunities for higher education and research so that the beneficiaries may not be at a disadvantage in the international level. However, many HEIs still encounter barriers that can affect their growth. The purpose of this study is to identify and interpret the most significant barriers affecting the growth of HEIs in India and understand the mutual interactions and interdependence among them.

Design/methodology/approach – In this study, 12 barriers affecting the growth of HEIs in India were identified and sorted by literature review, and these were validated through discussion with experts. Contextual relationships were drawn between them during brainstorming sessions. A hierarchical, seven-level structural model of these barriers affecting the growth of HEIs was developed by utilizing the total interpretive structural modeling (TISM) approach.

Findings – The results of this study showed “lack of climate change research” as the topmost outcome of the model having the highest dependence and lowest driving power. “Facilitator domination over academic experts” occupied the bottom-level input, compared to other barriers of the model, with highest driving power and lowest dependence. Further, Matrice d’Impacts Croisés – Multiplication Appliquée à un Classement (MICMAC) analysis was also performed to categorize the identified barriers for better understanding. Under MICMAC analysis, five barriers were categorized as driver barriers, four barriers as dependent barriers and the rest as linkage barriers, while none as autonomous. A TISM-based model that partitioned the barriers into different levels was developed, showcasing direct and important transitive relations.

Practical implications – The TISM-based model developed in this study may provide a more realistic approach to the problems faced by decision-makers, academicians and top managers in the growth of HEIs in India. Thus, it will provide a roadmap to focus on the most influential barriers and reduce or eliminate them.

Originality/value – The present study identified major barriers that when dealt with can help overcome the effect of other barriers preventing the growth of HEIs in India. It also introduces a model to identify and manage the barriers by understanding the effectiveness of each barrier and minimizing the overall effect in Indian HEIs. Moreover, the analysis of interdependence and interactions studied among these barriers may also help decision-makers, academicians and top managers to find deeper insights, prioritize and rectify them, to re-establish the growth of HEIs and gain a competitive advantage.

Keywords India, Barriers, Total interpretive structural modeling (TISM), Higher education institutions (HEIs)

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1. Introduction

Until the late 1970s, the higher education system in India was limited to primary education governed by a public-sector industry that was charging less or no fees ([Sahu et al., 2017](#)). The quality of higher education had not received enough attention until the late 1990s as two-thirds of the colleges and universities in India do not meet the required quality standards ([Dukkipati, 2010](#)). Advocates of higher education institutions (HEIs) assert that quality education is the key to holding an important place in the education industry globally. Ensuring quality education will also lead to better employability and aid in the country's development.

By the twenty-first century, the Government of India (GoI) began to concentrate more on the education sector. Foreign direct investment (FDI) of 100 percent was allotted to the sector, in addition to a considerable inflow of investment from the years 2000 to 2018 ([IBEF, 2019](#)). GoI has taken major steps to improve the quality standards of HEIs such as the National Accreditation Regulatory Authority Bill for Higher Education, Foreign Educational Institution Bill, Innovation Cell and Atal Ranking of Institutions and Innovation Achievements (ARIIA), Unnat Bharat Abhiyan, Skill Acquisition and Knowledge Awareness for Livelihood Promotion (SANKALP), Skill Strengthening for Industrial Value Enhancement (STRIVE) and many others to assess innovation efforts and encourage healthy competition among HEIs in the country ([IBEF, 2019](#)).

However, experts from HEIs further claim that lack of climate change research ([Filho et al., 2018a](#)), political pressure over academic environment, lack of exposure to academic environment ([Porter et al., 2016](#)) and paucity of funds ([Ramjeawon and Rowley, 2017](#)) are a hindrance to the expansion of higher education in India. This is contrary to the expectations of major stakeholders of the education system, because there is a positive association between the growth of HEIs and the development of a country ([Valero and Reenen, 2019](#); [Aghion et al., 2009](#)).

The extant literature further identified a number of impediments toward the growth of HEIs, but none has explored the most significant obstacles affecting the performance of HEIs of India. It is, therefore, imperative to identify the major barriers to the growth of HEIs, and if studied, can facilitate overcoming or lowering their effects on the academic environment. Therefore, this study is an attempt to identify the most important barriers preventing the growth of HEIs in India. Consequently, there is a need for a systematic approach to evaluate the barriers and ensure that once identified, they can be dealt with. The study also tries to develop an interrelationship framework of barriers.

This study has identified 12 major barriers through literature and brainstorming with experts from academia. For analyzing interrelationships among these, total interpretive structural modeling (TISM) and Matrice d'Impacts Croisés – Multiplication Appliquée à un Classement (MICMAC) analysis have been utilized.

The main objectives of this study are as follows:

- (1) to identify the major barriers preventing the growth of HEIs in India;
- (2) to establish a mutual relationship, relative importance and interdependence of each barrier with the help of TISM technique; and
- (3) to examine the driving power and dependence of the barriers affecting the HEIs' growth using MICMAC analysis.

The rest of the paper is organized as follows. [Section 2](#) presents literature reviews to identify the barriers and is summarized in a tabular format. [Section 3](#) discusses the research methodology used in this study. Results and discussions form [Section 4](#), followed by conclusion and future scope of the study.

2. Literature review

A large body of literature is available that demonstrates different barriers in the higher education system. Mazhar *et al.* (2014) and Wright and Horst (2013) had identified and described barriers capable of negatively impacting the higher education system. Pienaar and Bester (2016) explored the barriers that contributed to high levels of work stress among academic staff impacting job satisfaction, involvement and engagement. They identified 13 barriers that were responsible for creating poor work environment in HEIs. Chakraborty *et al.* (2016) identified and validated the barriers faced in incorporating sustainability in the curriculum of various higher education courses and classified them into three groups: lack of awareness, structure of education and lack of resources. Porter *et al.* (2016) studied the barriers to adoption of blended learning in higher education and developed a framework consisting of three stages – awareness/exploration, adoption and maturity implementation. The framework also identified the key strategy, structure and support issues that universities may address at different stages.

Sahu *et al.* (2017) presented issues related to gender inequalities in higher education. Financial constraints, lack of safety for women in public space, gender bias, gossip and social control within the family and the local community were the major barriers identified for young women's access to higher education. In another study, Garg *et al.* (2015) explored the most significant barriers to the unsuccessful implementation of information technology (IT) in HEIs and identified 14 barriers. Bilgohan *et al.* (2011) reported the common barriers within the higher education sector, including lack of resources, implementation cost, limited budget, lack of knowledge and skills, security and privacy concerns, technical issues, lack of trained staff, poor leadership, lack of culture change, time pressure, organization structure, lack of management vision, lack of planning and lack of perceived return on investments. Zakaraya and Zahir (2005) and Edington and Shin (2006) also presented similar barriers in the higher education sector. Aleixo *et al.* (2016) studied the barriers to sustainability in HEIs and identified six of them. They studied ambiguity and complexity of the concepts; lack of financial resources; resistance to change; organizational rigidity of the structure; lack of commitment, engagement, interest and involvement of faculty, students, staff, management and policymakers; and lack of training and specialization. Adams (2013); Waas *et al.* (2012); Versulst and Lambrechts (2015); Weber and Dudarstadt (2012) also highlighted similar barriers in HEIs.

Furthermore, Olsen *et al.* (2013), Yeh (2015), Barnett (2001), Moss *et al.* (2010) have also presented similar findings on HEIs. Antony *et al.* (2012) evaluated the adaptability of Lean Six Sigma (LSS) business improvement methodology for improving the efficiency and effectiveness of HEIs. They observed the challenges and barriers for the institution and development of LSS, namely, strategic objectives, lack of funds, lack of communication, cultural change and lack of vision of management and process management. Telsang and Kulkarni (2011) identified barriers like lack of academic audit and absence of faculty competency, to map implementation practices in engineering colleges for enhanced teaching–learning effectiveness. Liyange *et al.* (2018) reported the findings of a research study on the barriers to research and innovation in disaster resilience in HEIs in Asia. They revealed that administrative position, lack of management vision, lack of fund and domination of money and publicity over research learning, poor training and development of staff, not providing incentives to staff and reduced awareness on research and innovation initiatives were the major barriers in improving the research and innovation performance in HEIs. A recent study by Filho *et al.* (2018b) was on the role of transformation in learning and education for sustainable development in higher education. Their study showed that to enhance quality education, research education must acknowledge the value of transformation that offers the opportunity for researching and re-thinking about successful educational practices. The study further proposed the development of collaborative approaches by

Table I.
Barriers to HEIs' growth

Representation	HEI barrier	Identified studies
B1	Lack of management vision	Pienaar and Bester (2016); Liyanage <i>et al.</i> (2018); Antony <i>et al.</i> (2012)
B2	Facilitator domination over academic experts	Recommended by group of experts
B3	Curriculum as a classroom affair	Filho <i>et al.</i> (2018b); Von Blottnitz <i>et al.</i> (2015); Mulder <i>et al.</i> (2012); Dymont and Hill (2015)
B4	Lack of exposure to academic environment	Christo-Baker (2004); Humbert (2007); Porter <i>et al.</i> (2016);
B5	Lack of academic autonomy and self-respect	Recommended by group of experts
B6	Domination of money and public profile over teaching and learning	Liyanage <i>et al.</i> (2018)
B7	Lack of illustration of theories application	Hutchinson and Johnston (2004); Hart <i>et al.</i> (2008); Hewitt-Taylor <i>et al.</i> (2012)
B8	Lobbying for administrative positions	Liyanage <i>et al.</i> (2018); Lowery (2013)
B9	Poor structure of education	Porter <i>et al.</i> (2016); Humbert (2007)
B10	Lack of financial resources	Ramjeawon and Rowley (2017), Christo-Baker (2004); Humbert (2007); Porter <i>et al.</i> (2016); Liyanage <i>et al.</i> (2018);
B11	No academic audit	Liyanage <i>et al.</i> (2018); Telsang and Kulkarni (2011)
B12	Lack of climate change research	Christo-Baker (2004); Humbert (2007), Porter <i>et al.</i> (2016); Filho <i>et al.</i> (2018a);

academicians; re-designing of existing disciplines and curriculum, not a classroom affair; development of multicultural vision in academia; and reflection of the academics on the teachers as critical to foster sustainable learning in higher education.

The extant literature provides many barriers that hinder the growth of HEIs and other educational organizations, but none has identified the major barriers that can considerably impact the growth of HEIs and as a result, drive other issues. Table I summarizes the literature review of barriers to HEIs through literature survey and discussion with experts from academia.

3. Research methodology

The purpose of this study is to gain insight into the major barriers to the growth of HEIs in India and to understand the interrelationship between them. The research approach adopted in this study is based on TISM, which examines the contextual relationships among identified major barriers to HEIs growth. TISM is a novel extension of interpretive structural modeling (ISM) (Sushil, 2012). While ISM helps to understand the “what” and “how” of research (Warfield, 1974; Sage, 1977), TISM further allows a third question “why,” to validate the correctness of the model (Sushil, 2017). TISM is a better model to structure factors and to understand their interrelationship (Sushil, 2012; Sindhwani *et al.*, 2019). TISM is a modified and innovative modeling technique that generates a hierarchical model of the factors selected for the study, along with interpretation of each factor by an interactive process. It also displays a digraph that systematically depicts relationships among the factors (Hasan *et al.*, 2019). TISM is not only meant for developing a fully interpretive structural model but also helps in developing a knowledge base of all these interpretations and interrelations (Sindhwani *et al.*, 2019; Sandbhor and Botre, 2014). TISM is a powerful technique that has been employed by researchers in a number of applications that can be seen in the literature, presented in Table II.

Table II.
Application of TISM

Area of application	Reference
Modeling of barriers for lean green agile manufacturing system implementation	Sindhvani et al. (2019)
Modeling of barriers for lean implementation	Chaple et al. (2018)
Modeling the barriers of integrated sustainable solid waste management	Gopal et al. (2018)
Modeling of critical success factors of blockchain-based cloud services	Prasad et al. (2018)
Relational analysis of technological capabilities and supply chain resilience of firms	Rajesh (2017)
Framework for explaining sustainable supply chain performance	Shibin et al. (2017)
Modeling the enablers and barriers of flexible green supply chain management	Shibin et al. (2016)
Modeling of Critical Success Factors (CSFs) of smart phone manufacturing ecosystem	Jena et al. (2016)
Building theory of green supply chain management	Dubey et al. (2015a)
Generation of a sustainable manufacturing framework	Dubey et al. (2015b)
Modelling the enablers of total quality management (TQM) in airline	Singh and Sushil (2013)

Some of the multiple advantages of TISM are as follows ([Jena et al., 2016](#); [Prasad et al., 2018](#); [Chaple et al., 2018](#); [Srivastava and Sushil, 2013](#); [Sushil et al., 2018](#); [Hasan et al., 2019](#); [Shibin et al., 2017](#)):

- (1) interprets the links by explaining in what way barrier X will affect barrier Y;
- (2) identifies both direct and indirect transitive relationships among barriers;
- (3) digraphs simplify decision making and can be used on wider domains
- (4) processes both paired comparisons and transitivity checks simultaneously, thereby minimizing the redundancy in comparisons.

The overall methodology used in this research study is shown in [Figure 1](#). Initially, 16 barriers were identified through an extensive literature survey. Area experts were approached for a final list of barriers, and based on their opinion, 12 major barriers were selected for the current study.

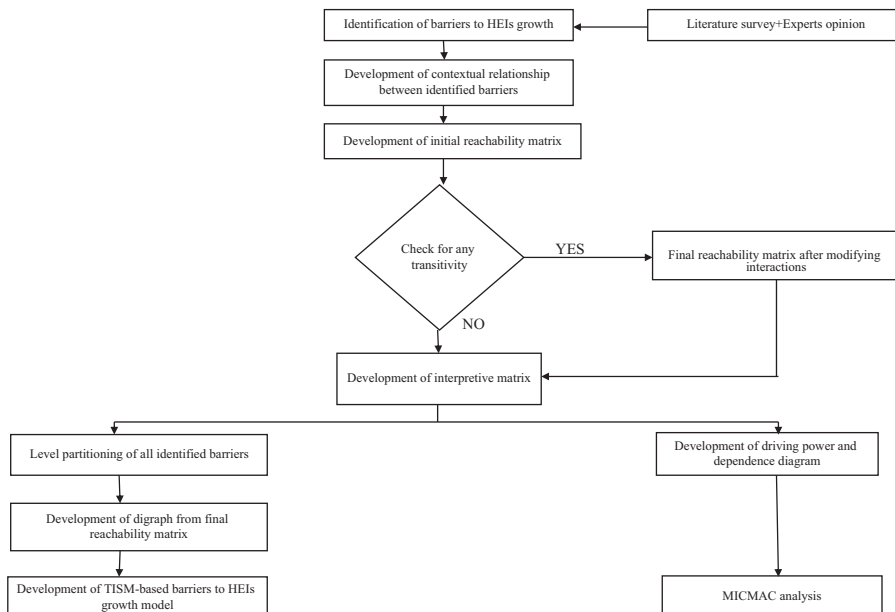


Figure 1.
Development of TISM

The second phase consisted of developing a contextual relationship between the barriers, which was then transformed into binary numbers to help develop an initial reachability matrix. The binary numbers 0 and 1 have the usual meaning, as represented in the ISM approach, where “0” stands for no relationship and “1” shows relationship between two barriers (Talib *et al.*, 2011a,b).

The next step was to check transitivity. If transitivity existed between any pair of barriers, the earlier relationships between the barriers impacting HEIs could be modified. The final reachability matrix was obtained after checking transitivity and modifying the necessary links.

Subsequently, an interpretive matrix was obtained where relationships in the contextual matrix were interpreted based on the understanding and knowledge of experts. This matrix will help the top management of HEIs to gain knowledge and understanding of major barriers and to remove them. In subsequent iterations, levels of barriers to HEIs’ growth are obtained using the level partition process.

The steps involved in developing TISM are as follows (Sushil, 2012; Sindhwani *et al.*, 2019; Yeh, 2015; Prasad *et al.*, 2018):

3.1 Step 1: identification of barriers to HEIs

This step includes identification of the barriers that require modeling of their interrelationship. For this, the barriers to HEI growth are identified from literature and discussion with experts from academia, as presented in Table I earlier.

3.2 Step2: contextual relationship development

The objective of this study was to identify and develop a contextual relationship between the barriers. ISM methodology suggests employing expert opinion based on various management techniques like the Delphi method, brainstorming and nominal techniques, to develop contextual relationships among barriers (Talib and Rahman, 2017; Kumar and Talib, 2017; Al-Esmael *et al.*, 2019). Thus, experts were approached to identify the contextual relationship among the barriers, as mentioned above. In total, 132 comparisons were made from the 12 major barriers identified. The four symbols as used in ISM – V, A, X and O – were chosen to develop relationships and their direction between the barriers (Talib *et al.*, 2011a; Faisal *et al.*, 2019). Consequently, the structural self-interaction matrix (SSIM) was produced, as shown in Table III.

3.3 Step3: initial reachability matrix

In this step, SSIM was converted into a binary initial reachability matrix, where 1 or 0 replaced the cells representing V, A, X and O based on the relationship, as explained by Talib *et al.* (2011a,b). The initial reachability matrix is shown in Table IV by following the substitution rule (Talib and Rahman, 2017; Faisal *et al.*, 2019; Talib *et al.*, 2011b).

3.4 Step4: transitivity check and final reachability matrix

The initial reachability matrix was subjected to a transitivity check, implying that the interpretations of relationships developed were examined to confirm if all the relations are correctly assessed (Chaple *et al.*, 2018). For example, if barrier B_i led to barrier B_j ($B_i \text{---} B_j$) and barrier B_j led to barrier B_k ($B_j \text{---} B_k$), then barrier B_i must lead to barrier B_k ($B_i \text{---} B_k$) where $i, j, k = 1, 2, 3, \dots, 12$. The process of overcoming such gaps is known as a transitivity check (Talib *et al.*, 2011a,b). The gaps that are listed in Table IV (initial reachability matrix) were examined and modified by incorporating the transitivity relations. The final reachability matrix is shown in Table V wherein transitivity is marked as 1*.

Table III.
SSIM

Barrier no.	Barrier name	12	11	10	9	8	7	6	5	4	3	2	1
1.	Lack of management vision	V	O	V	V	V	V	X	O	V	V	A	-
2.	Facilitator domination over academic experts	V	V	V	V	V	V	V	V	V	V	-	
3.	Curriculum as a classroom affair	V	X	A	A	A	V	O	A	A	-		
4.	Lack of exposure to academic environment	V	O	X	A	A	V	O	X	-			
5.	Lack of academic autonomy and self-respect	V	V	X	O	A	V	A	-				
6.	Domination of money and public profile over teaching and learning	V	V	O	O	V	V	-					
7.	Lack of illustration of theories application	V	A	O	A	O	-						
8.	Lobbying for administrative positions	V	V	O	X	-							
9.	Poor structure of education	V	V	V	-								
10.	Lack of financial resources	V	V	-									
11.	No academic audit	V	-										
12.	Lack of climate change research	-											

Note(s): Barrier No.: V: Variable i will help achieve variable j ; A: Variable j will be achieved by variable i ; X: Variable i and j will help achieve each other; and O: Variables i and j are unrelated

3.5 Step5: interpretive logic-knowledge base matrix

The value “1” represented in Table V (final reachability matrix) indicated the presence of a relationship between two major barriers and hence required interpretation. For this process, the opinion of experts was sought again. Cells having a value of “1” in Table V were replaced with the corresponding relationship as per their interpretation. An interpretive matrix evolved as shown in Table VI. For example, the value “1” in the final reachability matrix between “lack of management vision (B1)” and “lack of academic autonomy and self-respect (B5)” indicated “lack of management vision impedes self-respect and academic performance”

Table IV.
Initial reachability
matrix

Barrier no.	Barrier name	1	2	3	4	5	6	7	8	9	10	11	12
1.	Lack of management vision	1	0	1	1	0	1	1	1	1	1	0	1
2.	Facilitator domination over academic experts	1	1	1	1	1	1	1	1	1	1	1	1
3.	Curriculum as a classroom affair	0	0	1	0	0	0	1	0	0	0	1	1
4.	Lack of exposure to academic environment	0	0	1	1	1	0	1	0	0	1	0	1
5.	Lack of academic autonomy and self-respect	0	0	1	1	1	0	1	0	0	1	1	1
6.	Domination of money and public profile over teaching and learning	1	0	0	0	1	1	1	1	0	0	1	1
7.	Lack of illustration of theories application	0	0	0	0	0	0	1	0	0	0	0	1
8.	Lobbying for administrative positions	0	0	1	1	1	0	0	1	1	0	1	1
9.	Poor structure of education	0	0	1	1	0	0	1	1	1	1	1	1
10.	Lack of financial resources	0	0	1	1	1	0	0	0	0	1	1	1
11.	No academic audit	0	0	1	0	0	0	1	0	0	0	1	1
12.	Lack of climate change research	0	0	0	0	0	0	0	0	0	0	0	1

Table V.
Final reachability
matrix

Barrier no.	Barrier name	1	2	3	4	5	6	7	8	9	10	11	12	Driving power	Rank
1.	Lack of management vision	1	0	1	1	1*	1	1	1	1	1	1*	1	11	II
2.	Facilitator domination over academic experts	1	1	1	1	1	1	1	1	1	1	1	1	12	I
3.	Curriculum as a classroom affair	0	0	1	0	0	0	1	0	0	0	1	1	04	VI
4.	Lack of exposure to academic environment	0	0	1	1	1	0	1	0	0	1	1*	1	07	V
5.	Lack of academic autonomy and self-respect	0	0	1	1	1	0	1	0	0	1	1	1	07	V
6.	Domination of money and public profile over teaching and learning	1	0	1*	0	1	1	1	1	0	1*	1	1	09	III
7.	Lack of illustration of theories application	0	0	0	0	0	0	1	0	0	0	0	1	02	VII
8.	Lobbying for administrative positions	0	0	1	1	1	0	1*	1	1	0	1	1	08	IV
9.	Poor structure of education	0	0	1	1	1*	0	1	1	1	1	1	1	09	III
10.	Lack of financial resources	0	0	1	1	1	0	1*	0	0	1	1	1	07	V
11.	No academic audit	0	0	1	0	0	0	0	0	0	0	0	1	04	VI
12.	Lack of climate change research	0	0	0	0	0	0	0	0	0	0	0	1	01	VIII
	Dependence Power	03	01	10	07	08	03	11	05	04	07	10	12	81/81	
	Rank	VI	VII	III	V	IV	VI	II	V	VI	V	III	I		
Note(s): 1*entries are included to incorporate transitivity															

Barrier no.	B1	B2	B3	B4	B5	B6
B1	–	–	Curriculum is considered a classroom affair by management	Poor vision of management leads to poor academic environment	Lack of management self-respect and academic performance	Compromise on academic quality and standards
B2	Inefficient management system	–	Other incentives gained priority than academic	Political interference in academic	Disincentive for true academicians	Political interference in appointment
B3	–	–	–	–	–	–
B4	–	–	Insufficient training and related education	–	Lack of vision and enriched academic environment	–
B5	–	–	Lack of interest toward quality teaching and learning	Insufficient training and learning ability	–	–
B6	Facilitators dominating academics	–	Lack of vision toward education	–	Lack of interest toward learning and education	–
B7	–	–	–	–	–	–
B8	–	–	Political interference in academics	Political interference in appointments	Priority given to money and public profile	–
B9	–	–	Management unwillingness toward academics	Lack of institution vision	Lack of vision toward the academic performance	–
B10	–	–	Diminishing standards of education	No enrichment toward academic environment	Compromise toward academic quality	–
B11	–	–	Poor performance and outcome	–	–	–
B12	–	–	–	–	–	–

Barrier no.	B7	B8	B9	B10	B11	B12
B1	Lack of practical approach	Interest toward power and position	Management unwillingness toward academics	Management inability to visualize the short- and long-term goals	–	Lack of broad thinking toward advancement of research
B2	Political interference in academics	Political interference in administrative positions	Influence of political forces on academic system	Emergence of shortage of funds and irregularities	Political interference in information and evaluation system	–
B3	Lack of academic performance	–	–	–	Poor academic performance evaluation system	Lack of management attitude toward innovative learning and research

(continued)

Table VI.
Interpretive matrix for
TISM of barriers to
HEIs' growth

Table VI.

Barrier no.	B7	B8	B9	B10	B11	B12
B4	Inability to convert ideas into applications	-	-	Inability to generate fund, write projects and develop resources	Poor academic assessment system	Lack of vision toward academic environment
B5	Lack of interest toward practical approach	-	-	Lack of interest toward the institution development	Poor evaluation system	Lack of recognition from management
B6	Lack of interest toward teaching and learning	Interest toward power and position	-	Shifting of priority from education enhancement to power enhancement	Influence of money and power over academic evaluation	Influence of money and power over research environment
B7	-	-	-	-	-	Insufficient training on new and advance tools and techniques
B8	Position priority rather than academics	-	Priority given to position rather than education system	-	Political influence on top position	Priority given to position rather research and development
B9	Lack of vision towards practical approach	Management unwillingness toward the academics	-	Lack of vision toward education system	Management unwillingness toward academic performance evaluation	Lack of vision toward development of out-of-box research
B10	Insufficient funds lead to lack of practical abilities	-	-	-	Compromise toward academic assessment	No enrichment toward new and innovative research
B11	Insufficient practical ability	-	-	-	-	Poor performance and ranking in academics
B12	-	-	-	-	-	-

in the interpretive matrix. The interpretive matrix showed new and significant knowledge that will help top management and educationalists to understand better "*how*" and "*why*" major barriers to the growth of HEIs affect each other. As the development of this interpretive logic-knowledge base matrix involved advice from experts, this matrix can support and guide the management and educationalists to take appropriate decisions for HEIs' growth.

3.6 Step6: level partitioning in the reachability matrix

Level partitioning in the reachability matrix is carried out as in ISM, by identifying the reachability sets and antecedent sets (Faisal and Talib, 2017; Talib *et al.*, 2011a,b). Similar barriers from both the interaction and antecedents set were assigned a top-level position in the hierarchy and excluded from analyses, as these would not have influenced the remaining barriers. This process was iterated until all levels of barriers had been identified as shown in Tables VII and VIII.

As shown in Table VII, the barrier B12 (lack of climate change research) holding the top-most level of the hierarchy was removed, followed by the next iteration to obtain the next level of barriers. These levels produced a digraph and the final model. The process was completed in seven iterations, giving seven levels in the hierarchy model as summarized in Table VIII. Barrier B12 (lack of climate change research) occupied the top-most level and barrier B2 (facilitator domination over academic experts) the lowest level of the hierarchy, while all other barriers sat at various intermediate levels.

3.7 Step7: development of digraph

The digraph in Figure 2 represents major barriers and direction of relationships between them by nodes and directed links. Each barrier was arranged in the digraph as per its level obtained from Step6, and the direction of the relationship was established as per the final

Barrier no.	Reachability set R(Bi)	Antecedent set A(Bi)	Intersection set R(Bi)∩A(Bi)	Level
1	B1, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12	B1, B2, B6	B1, B6	
2	B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12	B2	B2	
3	B3, B7, B11, B12	B1, B2, B3, B4, B5, B6, B8, B9, B10, B11	B3, B11	
4	B3, B4, B5, B7, B10, B11	B1, B2, B4, B5, B8, B9, B10, B12	B4, B5, B10	
5	B3, B4, B5, B7, B10, B11, B12	B1, B2, B4, B5, B6, B8, B9, B10	B4, B5, B10	
6	B1, B3, B5, B6, B7, B8, B10, B11, B12	B1, B2, B6	B1, B6	
7	B7, B12	B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11	B7	
8	B3, B4, B5, B7, B8, B9, B11, B12	B1, B2, B6, B8, B9,	B8, B9	
9	B3, B4, B5, B7, B8, B9, B10, B11, B12	B1, B2, B8, B9	B8, B9	
10	B3, B4, B5, B7, B10, B11, B12	B1, B2, B4, B5, B6, B9, B10	B4, B5, B10	
11	B3, B7, B11, B12	B1, B2, B3, B4, B5, B6, B8, B9, B10, B11	B3, B11	
12	B12	B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12	B12	I

Table VII.
Barriers to HEIs' growth level iteration *i*

Table VIII.
Barriers to HEIs’
growth level iterations

Barrier no.	Barrier name	Reachability set R(Bi)	Antecedent set A(Bi)	Intersection set R(Bi)∩A(Bi)	Level
B12	Lack of climate change research	B12	B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12	B12	I
B7	stration of theories application	B7	B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11	B7	II
B3	Curriculum as a classroom affair	B3, B11	B1, B2, B3, B4, B5, B6, B8, B9, B10, B11	B3, B11	III
B11	No academic audit	B3, B11	B1, B2, B3, B4, B5, B6, B8, B9, B10, B11	B3, B11	III
B4	Lack of exposure to academic environment	B4, B5, B10	B1, B2, B4, B5, B8, B9, B10	B4, B5, B10	IV
B5	Lack of academic autonomy and self-respect	B4, B5, B10	B1, B2, B4, B5, B8, B9, B10	B4, B5, B10	IV
B10	Lack of financial resources	B4, B5, B10	B1, B2, B4, B5, B8, B9, B10	B4, B5, B10	IV
B8	Lobbying for administrative positions	B8, B9	B1, B2, B6, B8, B9,	B8, B9	V
B9	Poor structure of education	B8, B9	B1, B2, B6, B8, B9,	B8, B9	V
B1	Lack of management vision	B1, B6	B1, B2, B6	B1, B6	VI
B6	Domination of money and public profile over teaching and learning	B1, B6	B1, B2, B6	B1, B6	VI
B2	Facilitator domination over academic experts	B2	B2	B2	VII

reachability matrix (Step5). Significant indirect relationships in the final digraph are represented with a dotted line.

3.8 Step8: interpretive matrix

The digraph from Step7 was converted into a binary interaction matrix, in which “1” indicated direct as well as transitive links between the barriers. Significant relationships among barriers were then mentioned in the interpretive matrix using an interpretive logic-knowledge base table (Table VI).

3.9 Step9: formation of TISM-based model for barriers to HEIs’ growth

With the final reachability matrix in Table V and information obtained from the interpretive matrix (Table VI), the TISM-based model for barriers to HEIs’ growth was developed, as depicted in Figure 3. While formulating this model, all the barriers were arranged in the order of their levels as per Table VIII. Accordingly, all the barriers were placed one below the other with B12 holding Level I placed at the top. Each arrow in the image represents the relationship between two barriers, and their interpretation is illustrated by respective links leading to the formation of TISM for barriers to HEIs’ growth. The interpretation of some of the important transitive relations is as follows:

- (1) “Domination of money and public profile over teaching and learning (B6)” – “lack of financial resources (B10)” – may be due to “shifting of priority from education enhancement to power enhancement,” and may act as a major hindrance to the growth of HEIs.

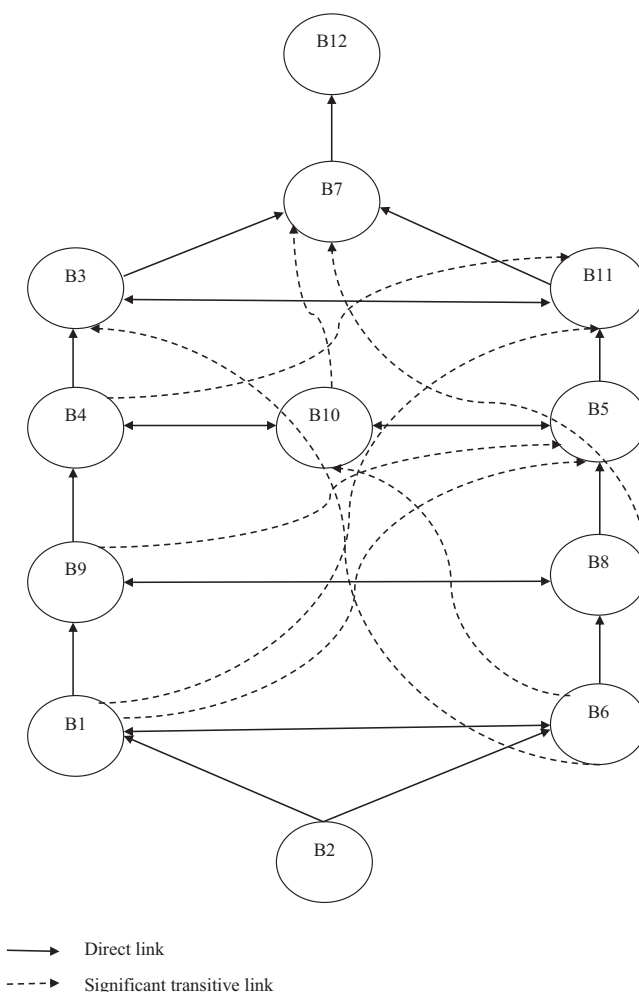


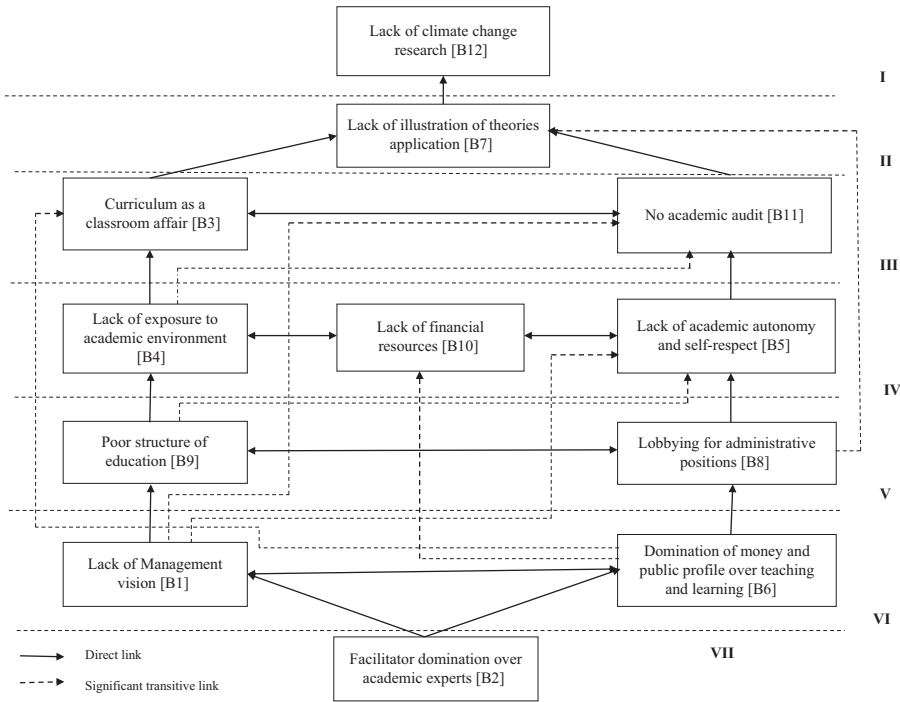
Figure 2.
Digraph showing
different levels

- (2) “Lack of management vision (B1)” – “lack of academic autonomy and self-respect (B5)” – this may be due to “lack of management vision impedes self-respect and academic performance.”
- (3) “Lack of exposure to the academic environment (B4)” – “No academic audit (B11)” – may be due to “poor academic assessment system.”
- (4) “Poor structure of education (B9)” – “Lack of academic autonomy and self-respect (B5)” – may be due to “lack of vision towards the academic performance.”

3.10 Step10: MICMAC analysis

IMICMAC analysis, popularly known as cross-impact matrix multiplication applied to classification, developed by [Duperrin and Godet \(1973\)](#), was used to analyze and classify the barriers of interest in terms of driving power and dependence ([Singh and Sushil, 2013](#)). Barriers impacting the growth of HEIs were classified into four different clusters based on

Figure 3.
TISM-based model for
barriers to HEIs'
growth



their driving power and dependence, where the row totals in the final reachability matrix represented the driving power and column totals indicated dependence (Table V). The driving power and dependence of each barrier was then plotted graphically and the driver power-dependence diagram was constructed, as shown in Figure 4. The graph was then divided into four clusters.

The first cluster is called "autonomous barriers cluster" and signifies weak driving power and weak dependence. The barriers in this cluster are relatively disconnected from other barriers to the system. In this study, no autonomous barriers were present, implying that all the identified barriers of the study were important, and therefore, all the 12 barriers selected have a significant impact on Indian HEIs growth.

The second cluster is known as "dependent barriers cluster" having strong dependence power but weak driving power. In this study, the following four barriers appeared within this cluster:

- (1) "Lack of climate change research (B12)" had high dependence power (12) and was placed at the top-most level in the TISM hierarchy.
- (2) "Lack of illustration of theories application (B7)" had weak driving power (02) and high dependence power (11), appearing at Level II of the TISM model.
- (3) "No academic audit (B11)" had weak driving power (4) and strong dependence power (10) and was thus placed at Level III of TISM.
- (4) "Curriculum as a classroom affair (B3)" had weak driving power (04) and high dependence power (10) and thereby also appeared at Level III of the TISM model.

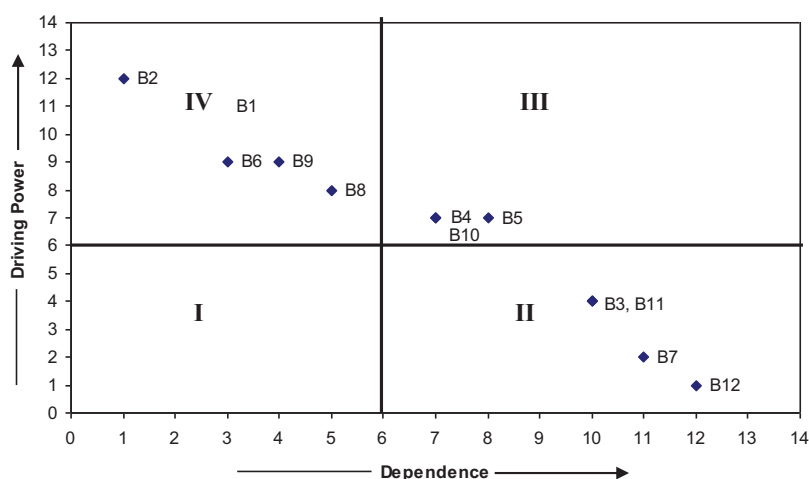


Figure 4.
MICMAC analysis of
barriers to HEIs'
growth

All the above barriers occupied the top levels of the TISM model due to their high dependency, indicating that they are barriers of high influence, and any changes on the other barriers can affect them.

"Linkage barriers cluster" is the third cluster that has both strong dependence power and strong driving power. These barriers are considered to be unstable. Any activity on these can have a significant effect on not just the other barriers but also themselves. In the current study, the following three barriers were identified as linkage barriers:

- (1) "Lack of exposure to academic environment (B4)" had high dependence (07) and high driving power (07) and appeared in the middle of the TISM model.
- (2) "Lack of financial resources (B10)" also had high dependence (07) and high driving power (07). Both the barriers were placed at Level IV of TISM.
- (3) "Lack of academic autonomy and self-respect (B5)" having strong driving power (07) and strong dependence power (08) occurred at Level IV of TISM.

The above linkage barriers were characterized as volatile and highly inter-correlated barriers due to their high dependence and driving power, occupying the middle of the TISM model.

The fourth cluster is called "driver barriers cluster." This cluster has strong driving power but weak dependence. Five barriers in this study were present in the cluster, as follows:

- (1) "Facilitator domination over academic experts (B2)" had strong driving power (12) and weak dependence power (01) receiving a position at the bottom level of TISM.
- (2) "Lack of management vision (B1)" had strong driving power (11) and weak dependence power (03) and occupied Level VI of the TISM model.
- (3) "Domination of money and public profit over teaching and learning (B6)" with strong driving power (09) and weak dependence (03) occupied Level VI of TISM.
- (4) "Poor structure of education (B9)" having strong driving power (09) and weak dependence power (04) was set at Level V of TISM.

- (5) "Lobbying for administrative positions (B8)" had strong driving power (08) and was also placed at Level V of TISM.

The above five barriers formed the base of the TISM model and had to be strategically handled as they can influence the other barriers. Hence, the study revealed that strategic and effective management of the driving and linkage barriers could control the dependence barriers like "lack of climate change research (B12)" of the higher education system of India.

4. Results and discussion

The objective of this study was to identify and model the barriers to the growth of Indian HEIs and further analyze the interactions among them. To achieve these objectives, the TISM-based model was developed so that academicians and top-management can consider these barriers for improved performance and growth of Indian HEIs. This will also certainly help Indian HEIs to improve their global performance, ranking and competition in the marketplace. While developing the TISM-based framework for barriers to HEIs' growth, as presented in Figure 3, both direct and transitive links among the barriers, defining their influence on each other, were taken into consideration. The direction of relationships was established based on the driving power and dependence power. Furthermore, an interrelationship was developed among the barriers by partitioning them into seven different levels as described in section 3.6, and represented in Figure 3. As shown, Level I is occupied by the barrier "lack of climate change research (B12)." Level II houses only one barrier – "lack of illustration of theories application (B7)." Barriers "curriculum as a classroom affair (B3)" and "no academic audit (B11)" fall in level III. Level IV consists of barriers "lack of exposure to the academic environment (B4)," "lack of financial resources (B10)" and "lack of academic autonomy and self-respect (B10)." Level V includes barriers "poor structure of education (B9)" and "lobbying for administrative positions (B8)". Barriers "lack of management vision (B1)" and "domination of money and public profile over teaching and learning (B6)" are held in Level VI. Lastly, barrier "facilitator domination over academic experts (B2)" falls at the bottom Level VII. The respective levels signify the relative positions occupied by the barriers in the TISM model. The barrier that receives the top-most place of the TISM model is the outcome of the other barriers. "Lack of climate change research (B12)" positioned at the pinnacle of the TISM hierarchy, is thus, dependent on mid- and bottom-level barriers. Hence, to improve the performance of Indian HEIs' growth, academicians and top management must focus on barriers having high driving power and should formulate strategies taking the help of the findings of this study. In the proposed model, "facilitators domination over academic experts (B2)" was identified as the most strategic barrier and could be key for the management of HEIs while designing an education system, as it can highly influence their performance. "Lack of management vision (B1)" and "domination of money and public profile over teaching and learning (B6)" can indirectly affect the academic quality, curriculum design, classroom teaching and learning ability. Further, barriers "poor structure of education (B9)" and "lobbying for administrative post (B8)" can also indirectly influence the poor academic environment and climate change research.

5. Conclusions

To conclude, the present study can be considered as an attempt to extend the available literature on barriers to HEIs' growth within the Indian context (Sahu *et al.*, 2017; Dukkipati, 2010; Chakraborty *et al.*, 2016; Garg *et al.*, 2015) and also contribute toward the existing

literature on TISM methodology (Singh and Sushil, 2013; Sushil, 2012; Jena *et al.*, 2016). The major contribution of this study can be considered under multiple headings:

- (1) This study has identified 12 major barriers that affect the growth of Indian HEIs.
- (2) The present work has developed a model that identifies the influence of these barriers, and their mutual relationship, relative importance and interdependence with the help of TISM and MICMAC analysis.
- (3) This study uses TISM model instead of ISM, as the latter cannot identify the mutual relationship between barriers.
- (4) This study also helps to answer the "*what*," "*how*" and "*why*" aspects by giving answers to the questions "*what are the barriers to Indian HEIs growth?*," "*why these barriers are linked in certain way?*" and "*how these barriers are interlinked?*"
- (5) This research identifies "facilitator domination over academic experts (B2)," "lack of management vision (B1)" and "domination of money and public profile over teaching and learning (B6)" as the major barriers that have a significant effect over the other linked barriers. These barriers have the strongest driving power and weakest dependence and lie at the bottom of the TISM model. "Lack of climate change research (B12)" with the lowest driving power and highest dependence emerges as an outcome barrier, highly influenced by the succeeding barriers. "Lack of exposure to the academic environment (B4)," "lack of financial resources (B10)" and "lack of academic autonomy and self-respect (B5)" are linking barriers that can influence other barriers as well as have a feedback effect on themselves.

Thus, this study tries to present the interrelationships and hierarchy of barriers to the growth of HEIs in a systematic way, by utilizing the extant literature survey in the relevant area, expert opinion and TISM methodology. These findings can help top management and academicians to prioritize and focus on those barriers that are impacting the growth and performance of HEIs.

Additionally, the results show that the involvement of management plays an important role in the higher education system, being responsible for the management of financial and administrative activities, including training and development activities of the staff, promotion of R&D and innovation opportunities, promotion and performance management, supplying support for research projects, encouraging entrepreneurship development and improvement of existing facilities and working conditions.

5.1 Managerial implications

In accordance with the TISM-based model for barriers to HEIs' growth, this study suggests some implications to managers for better performance and development in HEIs.

- (1) Identification of major barriers to the growth of HEIs will help academicians and top management to focus and analyze this area based on their impact and hierarchical level in the TISM model. With the model's support, academicians and administrators can concentrate on overcoming the barriers toward effective and improved performance of the Indian higher education system.
- (2) The study also calculated the driving power and dependence of all the 12 identified barriers based on their inter-linkage. This will help top management, educationalists and academicians to prioritize their efforts on these barriers in a systematic way, for achieving desired goals.

- (3) The barriers “lack of management vision (B1),” “facilitator domination over academic experts (B2),” “domination of money and public profile over teaching and learning (B6),” “poor structure of education (B9)” and “lobbying for administrative positions (B8)” are more strategic in nature and need to be handled carefully and smartly. Administrators and academicians should pay maximum attention to these for better performance of HEIs and long-term growth.
- (4) The growth and development of HEIs are dependent on academic staff and available infrastructure, which are primarily responsible for their functioning. The involvement of external participants such as politicians, facilitators and mentors will affect the academic and research environment in HEIs. Managers should understand that the success of HEIs is directly linked to the generation of funds, proper structure of education, management approach toward R&D and innovation activities for world-class research, application of theories to real-time work environment, regular quality certification and audit and training and development of staff.

5.2 Scope for future research

Although this study offers important aspects related to HEIs’ growth, the current research work can be extended by replacing MICMAC with the fuzzy MICMAC approach to improve the sensitivity. The study lacks a systematic questionnaire survey that could further test the model and provide relative weights to the barriers. Future studies can consider validating the model using structural equation modeling (SEM) and confirmatory factor analysis (CFA) using LISREL software. The tool can test the validity of such hypothetical models. This study identified 12 major barriers to HEIs’ growth with limited available literature in the area. Future studies can further explore, assess and validate other barriers to understand the problems and issues related to HEIs’ growth.

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About the authors

Faisal Talib is an Associate Professor at Department of Mechanical Engineering, Zakir Husain College of Engineering and Technology, Aligarh Muslim University (AMU), Aligarh (UP), India. He holds a PhD degree from the Indian Institute of Technology, Roorkee, and Masters in Industrial and Production Engineering from AMU. He has 23 years of teaching experience and has more than 90 publications to his credit in national/international journals and conferences. His special interests include quality engineering, TQM, service quality, quality concepts, industrial and production management, operations management and quality management in service industries. Many of his research papers are published in reputed journals like *Service Business (SBUS)*, *Total Quality Management and Business Excellence (TQMBE)*, *The TQM Journal*, *International Journal of Mobile Communication (IJMC)*, *Benchmarking: An International Journal (BIJ)*, *Journal of Systems and Information Technology (JSIT)*, *Journal of Foodservice Business Research (JFBR)*, *International Journal of Quality and Reliability Management (IJQRM)*, *Services Marketing Quarterly (SMQ)*, to name a few. Faisal Talib is the corresponding author and can be contacted at: ftalib77@gmail.com

Dr. Zillur Rahman is a Professor at the Department of Management Studies, IIT Roorkee. He is a recipient of the Emerald Literati Club Highly Commended Award and one of his papers was The Science Direct Top 25 Hottest Article. His work has been published and cited in various journals, including *Management Decision*, *Managing Service Quality*, *International Journal of Information Management*, *Industrial Management and Data Systems*, *The TQM Magazine*, *Business Process Management Journal*, *International Journal of Service Industry Management*, *Information Systems Journal*, *Decision Support Systems*, *Journal of Business and Industrial Marketing* and *International Journal of Computer Integrated Manufacturing*, to name a few.

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